SM90 FP4 MegaMoE | NVIDIA H200

82 workload configurations | NVFP4 fused, NVFP4 split, MXFP4 fused | current kernel latency

Cold L2 before each call; CUDA events inside a graph; per-call max across ranks, then median of 50 samples.

NVFP4 | fused + split

24 workloads / EP8

M / rank	Capacity	Kernel latency (us)
NVFP4 FUSED | Flash EP8		
1	384	**81.280**
24	384	**215.440**
32	384	**218.944**
64	384	**232.496**
96	384	**227.824**
NVFP4 FUSED | Mimo EP8		
1	384	**136.560**
24	384	**422.048**
32	384	**426.096**
64	384	**458.912**
96	384	**456.368**
NVFP4 FUSED | Pro EP8		
1	384	**161.856**
24	384	**703.008**
32	384	**706.304**
64	384	**727.424**
96	384	**746.960**
NVFP4 SPLIT | Flash EP8		
384	4224	**339.296**
1024	4224	**601.840**
4096	4224	**1,824.592**
NVFP4 SPLIT | Mimo EP8		
384	4224	**635.584**
1024	4224	**1,207.888**
4096	4224	**3,887.568**
NVFP4 SPLIT | Pro EP8		
384	4224	**949.056**
1024	4224	**1,622.128**
4096	4224	**6,017.504**

MXFP4 | fused

24 workloads / full token range

M / rank	Capacity	Kernel latency (us)
MXFP4 FUSED | Flash EP8		
1	4224	**97.296**
24	4224	**197.600**
32	4224	**192.016**
64	4224	**202.672**
96	4224	**220.752**
384	4224	**393.040**
1024	4224	**704.352**
4096	4224	**2,406.736**
MXFP4 FUSED | Mimo EP8		
1	4224	**127.344**
24	4224	**339.296**
32	4224	**360.208**
64	4224	**392.960**
96	4224	**434.160**
384	4224	**647.120**
1024	4224	**1,355.680**
4096	4224	**Pending***
MXFP4 FUSED | Pro EP8		
1	4224	**157.376**
24	4224	**519.264**
32	4224	**550.400**
64	4224	**614.960**
96	4224	**637.424**
384	4224	**750.672**
1024	4224	**1,810.048**
4096	4224	**29,917.920**

MXFP4 | fused

34 workloads / EP8 + EP4

M / rank	Capacity	Kernel latency (us)
MXFP4 FUSED | Flash EP8		
8	384	**171.856**
16	384	**182.560**
32	384	**193.280**
48	384	**197.504**
64	384	**202.880**
96	384	**220.672**
112	384	**221.040**
128	384	**221.904**
160	384	**236.144**
192	384	**229.760**
256	384	**1,060.400**
384	384	**1,155.168**
MXFP4 FUSED | Flash EP4		
8	768	**211.152**
16	768	**267.680**
32	768	**286.512**
48	768	**306.160**
64	768	**330.000**
96	768	**340.400**
128	768	**351.376**
160	768	**384.320**
192	768	**386.256**
224	768	**386.784**
256	768	**388.768**
384	768	**394.448**
512	768	**406.992**
768	768	**694.912**
MXFP4 FUSED | Mimo EP4		
8	384	**361.936**
32	384	**549.344**
64	384	**647.968**
128	384	**711.840**
192	384	**791.616**
240	384	**795.072**
288	384	**797.056**
384	384	**803.552**

Times in microseconds. Capacity = allocated maximum tokens per rank. NVFP4 split timing includes both L1 and L2.

Flash: H4096 / I2048 / E256 / K6 | MiMo: H6144 / I2048 / E384 / K8 | Pro: H7168 / I3072 / E384 / K6

* MiMo EP8 M=4096: timing pending verification. Observed batches: 14,712.832 / 4,816.128 / 5,118.064 us; raw measurements retained in CSV.